

A PROJECT Title
ON
ADAPTIVE NUTRITION PLANNING PLATFORM FOR SMART
INPATIENT HEALTHCARE

Insem-End project submitted in partial fulfilment of the requirements for the Course

Front End Development Frameworks *for the degree of*

BACHELOR OF TECHNOLOGY
IN
Computer Science and Engineering
(2025-2026 A.Y)

BY

M Ramya Sree **2520030094**

D Varun Teja Reddy **2520090221**

Under the Esteemed guidance of

Dr. D. Ramya Krishna

(Asst Prof, Dept of CSE)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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Name of Title : Adaptive Nutrition Planning Platform for Smart Inpatient Healthcare

ABSTRACT :

The *Adaptive Nutrition Planning Platform for Smart Inpatient Healthcare* is an application designed to improve the efficiency and effectiveness of meal planning in hospital environments. Proper nutrition plays a key role in patient recovery, and it requires careful consideration of factors such as dietary restrictions, allergies, medical conditions, and required calorie intake. Traditional methods of nutrition planning are often time-consuming, prone to errors, and lack proper coordination between healthcare staff and kitchen teams.

This platform provides an advanced and organized approach to hospital nutrition planning by considering all essential factors. It includes features such as patient dietary profile creation, intelligent meal planning, and automated kitchen order processing. The system helps in creating personalized nutrition plans for patients based on their medical conditions while ensuring that all dietary restrictions are properly followed. In addition, it supports storing historical dietary data, exporting information for analysis, and managing patient nutrition records efficiently.

The application also includes an administrative portal that allows easy monitoring of patient data and smooth handling of daily operations related to nutrition planning. It is built using modern technologies such as HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, and RESTful APIs, ensuring a responsive and user-friendly experience.

Overall, the platform helps reduce human errors in meal planning and improves coordination between healthcare professionals and kitchen staff. By automating important processes and supporting better decision-making, it contributes to improved patient care and more efficient hospital operations.

1. INTRODUCTION:

2. SOFTWARE REQUIREMENTS:

Course Coordinator

Course Instructor

FED , Coordinator